



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology

Department of Computer Science

CSC 3215: Web Technologies

Semester: Summer 25-26

Group No: 3

Section: G

Group Members:

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Project Title

House Rental Management System.

Short Description

The House Rental Management System is designed to manage the complete house rental process through four types of users: Admin, Manager, Owner, and Client. The system will allow owners to post houses for rent, while managers can review and approve or reject those house posts. Clients will be able to search and filter available houses, view house details, compare houses, and send booking requests. Owners will manage booking requests and rental information. The system will also provide payment tracking and notification features. The Admin will monitor users, houses, rental activities, and overall revenue and earnings through the dashboard.

Introduction

Finding and managing rental houses can be difficult when house information, booking requests, approvals, and payments are handled manually. A centralized House Rental Management System can make this process more organized and convenient for all users. This project aims to develop a system where Admin, Manager, Owner, and Client can perform their respective activities through a structured platform. In this system, owners can post houses for rent, and managers can approve or reject the submitted house posts. Once a house becomes available, clients can search and filter houses, view detailed information, compare different houses, and submit booking requests. Owners can accept or reject booking requests and manage active rentals. The system also supports rental and payment status tracking, payment notifications, and revenue monitoring. Through these features, the proposed system will provide a systematic and efficient way to manage house rental activities.

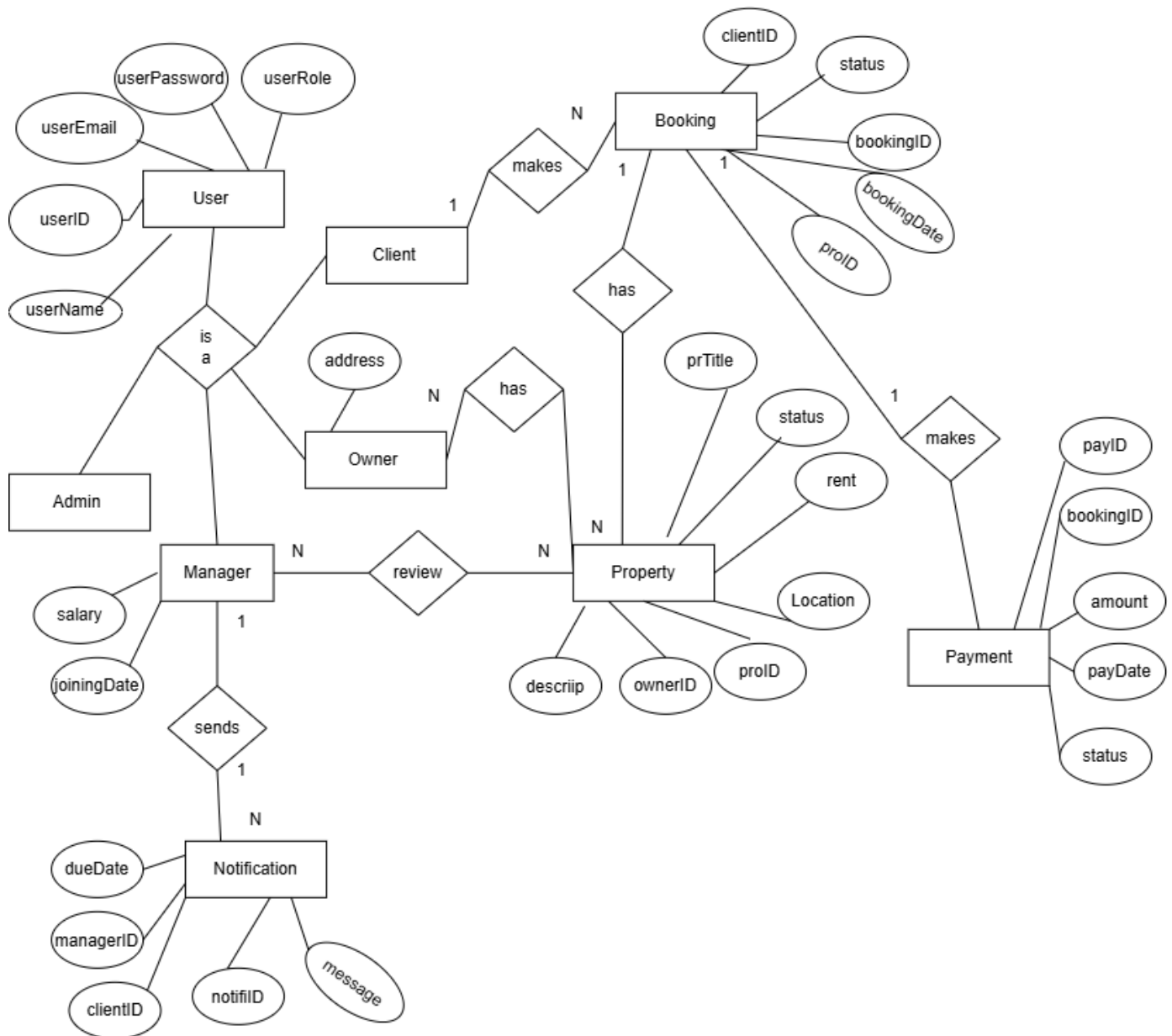
Background Study

House rental management involves several activities, including house listing, house approval, customer searching, booking, rental management, payment tracking, and communication between different users. In a traditional or less organized rental process, these activities may require separate communication and manual record keeping, which can make it difficult to track house availability, booking status, rental status, and payments.

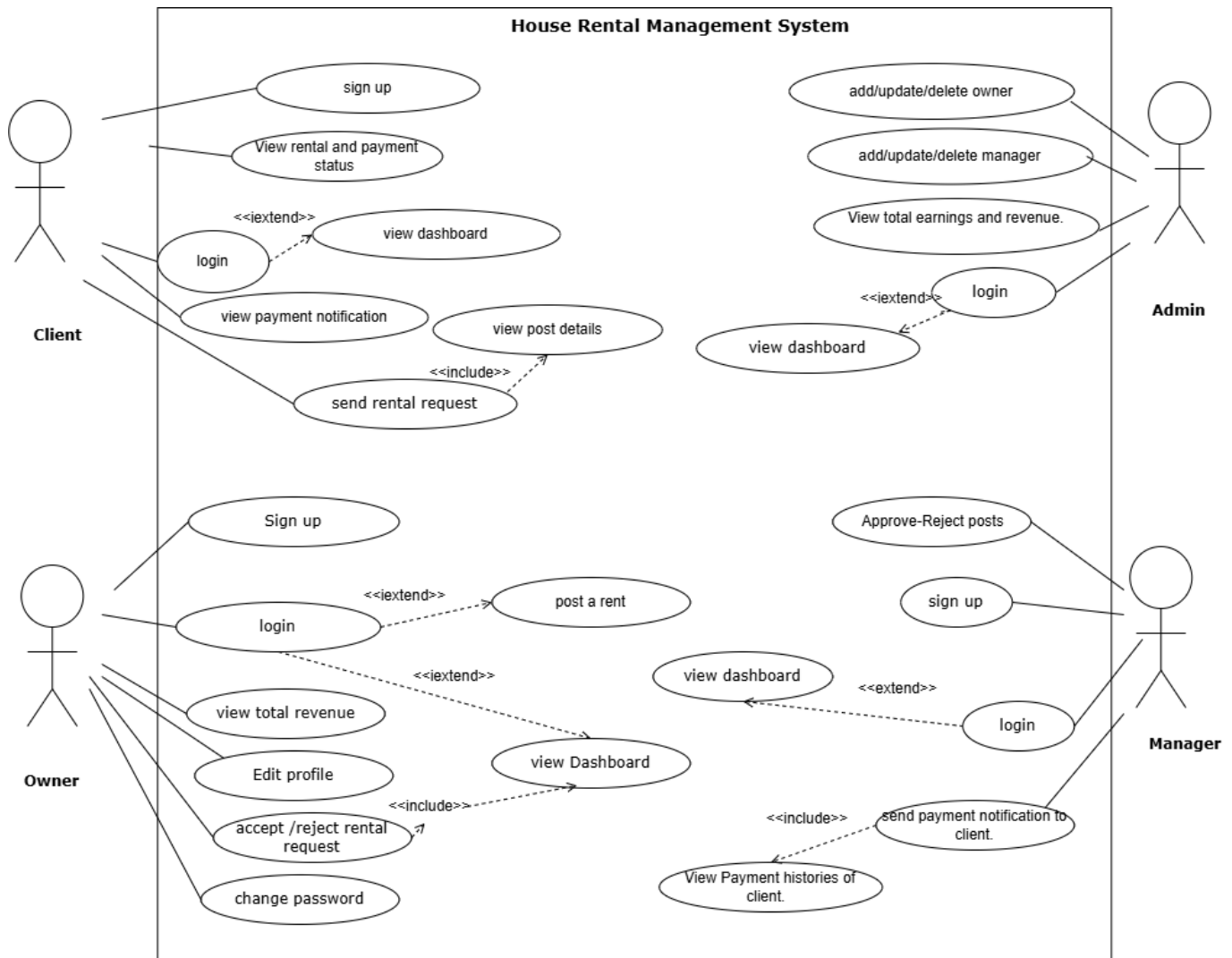
The proposed House Rental Management System organizes these activities into a single platform with role-based functionality. Owners can post houses and manage booking requests, while Managers can review house posts and

approve or reject them. Clients can search and filter available houses, view details, compare houses, and send booking requests. The system also provides payment tracking and notifications so that rental and payment information can be monitored more easily. Finally, the Admin can manage managers and owners and monitor total clients, houses, rentals, revenue, and earnings. Therefore, the project provides an organized workflow for managing the different stages of house rental..

ER Diagram



Use Case Diagram



UI (Figma/Sketch)

R

RentEase

Create your account

Find a place to rent, or list one of your own.

Full name

Jordan Smith

Email

name@company.com

Password

Create a password

I am a

Client

Owner

Already have an account? [Sign in](#)

Create account

R

RentEase

Sign in

Sign up

Email

name@company.com

Password

Enter your password

[Forgot password?](#)

Sign in

Don't have an account? [Sign up](#)

RentEase

ADMIN PORTAL

Dashboard

Managers

Owners

\$Earnings

AD

Admin Supervisor

admin@rentease.com

Dashboard Overview

Total Clients

128

Total Managers

12

Total Owners

45

Total Properties

230

Available Properties

142

Available

Rented Properties

88

Rented

Managers

Q Search listings, people, emails...

Name	Email	Phone	Status
☐ Robert Fox	robert.f@rentease.com	+1 (555) 019-2834	Active
☐ Cody Fisher	cody.f@rentease.com	+1 (555) 017-4839	Inactive
☐ Priya Patel	priya.p@rentease.com	+1 (555) 012-1100	Active
☐ Jordan Lee	jordan.l@rentease.com	+1 (555) 013-2299	Inactive
☐ Sofia Garcia	sofia.g@rentease.com	+1 (555) 018-4422	Active
☐ Daniel Kim	daniel.k@rentease.com	+1 (555) 014-5511	Inactive
☐ Maya Chen	maya.c@rentease.com	+1 (555) 015-6633	Active
☐ Omar Hassan	omar.h@rentease.com	+1 (555) 016-8844	Inactive
☐ Ava Nguyen	ava.n@rentease.com	+1 (555) 017-2200	Active
☐ Liam Brooks	liam.b@rentease.com	+1 (555) 013-7711	Inactive

New

Refresh

Delete

Save

Name

Email

Phone

Salary

Joining Date



Owners

Q Search listings, people, emails...

Name	Email	Phone	Status
☐ Robert Fox	robert.f@rentease.com	+1 (555) 019-2834	Active
☐ Cody Fisher	cody.f@rentease.com	+1 (555) 017-4839	Inactive
☐ Zoe Bennett	zoe.b@rentease.com	+1 (555) 017-5599	Active
☐ Lucas Ross	lucas.r@rentease.com	+1 (555) 014-7788	Inactive
☐ Mia Foster	mia.f@rentease.com	+1 (555) 015-2211	Active
☐ Logan Price	logan.p@rentease.com	+1 (555) 016-6644	Inactive
☐ Oliver Wright	oliver.w@rentease.com	+1 (555) 014-9900	Inactive

New

Refresh

Delete

Save

Name

Email

Phone

Address

Earnings & Revenue

Total Earnings

\$125,400

This Month

\$12,300

Pending Payments

\$3,200

Recent Transactions

Date	Property	Client	Amount	Status
Oct 24, 2024	Sunset Penthouse Apt 401	Alice Green	\$2,400	Paid
Oct 22, 2024	Oakridge Family Townhouse	Mark Specter	\$1,850	Paid
Oct 20, 2024	Beachfront Cabin Resort	Jessica Bell	\$3,100	Pending
Oct 18, 2024	Downtown Studio Flat	Dan Richardson	\$1,200	Paid
Oct 15, 2024	Suburban Villa Estates	Nancy Wheeler	\$4,500	Pending

Available Properties

Q Search location...

Location (All) ▾

Price Range ▾

Property Type ▾



Sunset Luxury Penthouse

Downtown Seattle, WA

3 Beds 2 Baths

\$2,800/mo

View Details



Oakwood Townhouse

Portland, OR

2 Beds 1.5 Baths

\$1,950/mo

View Details



Ocean Breeze Cabin

Malibu, CA

2 Beds 2 Baths

\$3,400/mo

View Details

RENT EASE

CLIENT PORTAL

Dashboard

My Rentals

Payments

Compare

Notifications

CL

Sarah Jenkins

sarah.j@mail.com

Property Insights



Sunset Luxury Penthouse

\$2,800/mo

1208 Pine Crest Way, Seattle, WA 98101

Bedrooms

Bathrooms

Area Size

Type

Available From

3 Beds

2 Baths

1,450 sqft

Penthouse

Nov 1, 2024

About this Property

Experience spectacular high-rise living with floor-to-ceiling windows offering sunset views of Seattle. This premium unit comes with high-end stainless steel appliances, premium wooden flooring, smart automated blinds, and a private balcony. Accessible to rapid transit, gourmet restaurants, and active downtown life.

Owner Contact

DM David Miller

Premium Listing Partner

david@millerholdings.com

+1 (555) 019-3958

Book Now

Compare Property

RENT EASE

CLIENT PORTAL

Dashboard

My Rentals

Payments

Compare

Notifications

SJ

Sarah Jenkins

sarah.j@mail.com

My Rentals & Payments

Active Rentals

2 Active Leases

Total Paid Transactions

\$8,400

Property Name	Address	Monthly Rent	Start Date	Lease Status	Payment
Sunset Luxury Penthouse	Downtown Seattle, WA	\$2,800/mo	Jan 01, 2025	Active	Paid
Oakwood Townhouse	Portland, OR	\$1,950/mo	Feb 15, 2025	Active	Due
Metropolitan Studio	New York, NY	\$1,600/mo	Jun 01, 2024	Expired	Paid
Malibu Beach Cabin	Malibu, CA	\$3,400/mo	Aug 10, 2024	Expired	Paid



RentEase
CLIENT PORTAL

Dashboard

My Rentals

Payments

Compare

Notifications

Compare Properties

Preview



Name	Sunset Luxury Penthouse	Ocean Breeze Cabin
Location	Downtown Seattle, WA	Malibu, CA
Price/Month	\$2,800/mo	\$3,400/mo
Bedrooms	3 Beds	2 Beds
Bathrooms	2 Baths	2 Baths
Area	1,850 sqft	1,400 sqft
Property Type	Penthouse Apartment	Beach Cabin
Availability	Available	Available

SJ

Sarah Jenkins
sarah.j@mail.com



RentEase
CLIENT PORTAL

Dashboard

My Rentals

Payments

Compare

Notifications

Payment Notifications

●

Rent Due Soon — Sunset Luxury Penthouse
Your monthly payment of \$2,800 is due in 3 days. Please review and process the transfer to avoid penalties.

10 mins ago

●

Payment Confirmed — Oakwood Townhouse
Successfully received rent payment of \$1,950 for the billing cycle of February 2025. Your receipt is now available.

2 hours ago

SJ

Sarah Jenkins
sarah.j@mail.com

Dashboard

Posts

Payments

Notifications

Dashboard Overview

Total Rental Posts

85 Posts

Accepted Posts

52 Accepted

Rejected Posts

12 Rejected

Pending Posts

21 Pending

Dashboard

Posts

Payments

Notifications

Rental Posts

All Posts (85)

Pending (21)

Accepted (52)

Rejected (12)

Property Name	Owner	Location	Price	Submitted	Status	Actions
Parkside Modern Villa	Alice Henderson	Seattle, WA	\$3,800	Today	Pending	<button>Approve</button> <button>Reject</button>
Sunset Luxury Penthouse	Sarah Jenkins	Seattle, WA	\$2,800	Jan 20, 2025	Accepted	No action required
Riverside Cabin Deluxe	Cody Fisher	Denver, CO	\$3,100	Jan 15, 2025	Rejected	No action required

Payment Histories

Search client name, property, transaction id...

Client Name	Property	Amount	Payment Date	Method	Status
Sarah Jenkins	Sunset Luxury Penthouse	\$2,800	Jan 01, 2025	Bank Transfer	Completed
Robert Fox	Oakwood Townhouse	\$1,950	Jan 15, 2025	Credit Card	Completed
Jenny Wilson	Metropolitan Studio	\$1,600	Jan 12, 2025	Bank Transfer	Pending
Cody Fisher	Malibu Beach Cabin	\$3,400	Jan 08, 2025	Credit Card	Failed
Sarah Jenkins	Oakwood Townhouse	\$1,950	Dec 15, 2024	Bank Transfer	Completed

Send Payment Notification

New Payment Notification

Select Client	Select Property
Mark Vance (Client #1092)	Sunset Villa #4B
Amount Due (\$)	Due Date
1200.00	2025-02-15
Notification Message	
Dear Mark, this is a friendly reminder that your monthly rent of \$1,200.00 for Sunset Villa #4B is due on February 15, 2025. Please process the payment through your client portal. Thanks!	

Send Notification

Recently Sent Notifications

Client	Property	Amount	Due Date	Sent Date	Status
Mark Vance	Sunset Villa #4B	\$1,200	Feb 15, 2025	Today, 09:30 AM	Sent
Samantha Green	Charming Urban Loft #12	\$2,100	Feb 10, 2025	Yesterday, 02:15 PM	Read
John Doe	Parkside Modern Villa #1	\$3,800	Feb 05, 2025	Jan 28, 2025	Read

Dashboard Overview

Total Rental Posts

5 Posts

My available rents

52 Rents

Rent Ongoing

12 Rents

Post a Rental

Enter Property Details

Property Title

e.g. Modern Lakefront Cabin with Private Dock

Property Type

Villa

Location / Address

742 Evergreen Terrace, Springfield

Monthly Rent (\$)

3200

Bedrooms

3

Bathrooms

2.5

Area (sqft)

1850

Availability Date

2025-03-01



Description

Describe the properties features, amenities, local neighborhood, school district and transit accessibility...

Property Images

[Click to upload images](#)

PNG, JPG or JPEG up to 10MB each

Post

RentEase

OWNER PORTAL

Dashboard

Post Rental

Bookings

Revenue

EM

Elena Mitchell

elena.m@rentease.com

Booking Requests

Client Name	Property	Requested Date	Duration	Message	Actions
Mark Vance	Downtown Penthouse Suite	Feb 12, 2025	12 months	Hoping to move in mid-Feb. Stable jo...	AcceptReject
Clara Oswald	Cozy Suburb Townhouse	Feb 01, 2025	6 months	Love the terrace garden. Ideal place f...	Accepted
Gregory Peck	Downtown Penthouse Suite	Jan 20, 2025	24 months	Requesting corporate lease agreeme...	Rejected
Jenny Wilson	Sunset Mountain Cabin	Mar 05, 2025	3 months	Weekend getaway lease request for t...	AcceptReject

RentEase

OWNER PORTAL

Dashboard

Post Rental

Bookings

Revenue

EM

Elena Mitchell

elena.m@rentease.com

Total Revenue

Total Revenue

\$45,200

This Month

\$4,800

Active Tenants

3

Payment Records

Property	Tenant	Amount	Date	Payment Method	Status
Cozy Suburb Townhouse	Clara Oswald	\$1,900.00	Feb 01, 2025	Direct Deposit	Received
Seaside Family Bungalow	Gregory Peck	\$4,500.00	Jan 28, 2025	Credit Card	Received
Sunset Mountain Cabin	Thomas Wright	\$2,450.00	Jan 15, 2025	PayPal	Received
Downtown Penthouse Suite	Mark Vance	\$4,200.00	Jan 05, 2025	Direct Deposit	Pending
Midtown Loft & Studio	Jenny Wilson	\$2,150.00	Jan 02, 2025	Direct Deposit	Received

Types of Users (Roles):

1. Admin
2. Manager
3. Client
4. Owner

Common Features (Available to All Types of Users)

- **Authentication**
 1. Login to the system
 2. Logout from the system
 3. User registration (Account Creation)
- **Account Management**
 1. Change and reset password
 2. Manage profile information (view, edit, delete, including profile picture)
- **Dashboard**
 1. Access a personalized dashboard after login (each type of user must land on their own dashboard after login)

Specific Features by User Type

- **Features of Admin:**
 1. **Manage Managers:** Add, update, or delete manager accounts.
 2. **Manage Owners:** Add, update, or delete owner accounts.
 3. **Earnings & Revenue:** Monitor the platform's total earnings and revenue.
- **Features of Manager:**
 1. **Manage Rental Posts:** approve or reject property rental posts.
 2. **View Payment Information:** View clients rental payment histories.
 3. **Send Payment Notification:** Send payment reminders or notifications to clients.
- **Features of Client:**
 1. **View Post Details:** View complete information about a selected property.
 2. **Rental & Payment Status:** Track current rental and payment information.
 3. **Property Comparison:** Compare between 2 properties to choose the most suitable one.
 4. **View Payment Notifications:** Receive notifications about payment.
- **Features of Owner:**
 1. **Post a Rent:** Create and publish a property rental post.
 2. **Booking Management:** Accept or reject booking requests from clients.
 3. **Revenue Tracking:** View the total revenue generated from rented properties.

N.B: Each user type must have at least 3 distinct features different from the others. Additional features may be added if desired. Proposals with generic, incomplete, or duplicated features across user types will receive reduced marks.